

KINGSTON CLIMATE, ON, Canada

WMO: 718200

Lat: 44.2232N

Lon: 76.5994W

Elev: 93

StdP: 100.21

Time Zone: -5.00 (NAE)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-22.0	-18.5	-26.7	0.3	-21.0	-23.7	0.4	-17.5	13.8	-2.1	12.4	-1.8	3.5	20	0.582

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.5	27.9	21.7	26.5	21.2	25.4	20.7	23.6	26.1	22.7	25.0	21.9	24.3	5.3	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.7	17.6	25.1	22.0	16.8	24.4	21.1	15.9	23.8	70.9	26.2	67.7	25.0	64.8	24.4	26.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
11.2	9.8	8.7	-26.8	31.1	3.1	1.9	-29.0	32.4	-30.8	33.5	-32.6	34.6	-34.8	35.9		
			-27.0	25.2	3.0	1.0	-29.2	25.9	-31.0	26.5	-32.7	27.1	-34.9	27.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	7.7	-6.7	-5.6	-0.8	5.8	13.1	17.3	21.0	20.3	16.4	9.8	3.3	-2.3
	DBStd	10.67	6.62	6.00	5.45	3.96	4.00	2.90	2.71	2.66	3.89	4.24	4.81	5.85
	HDD10.0	2083	517	436	335	136	17	0	0	0	2	56	204	381
	HDD18.3	4088	776	669	592	377	166	53	7	12	83	263	451	639
	CDD10.0	1248	0	0	2	8	114	219	341	318	192	51	3	0
	CDD18.3	213	0	0	0	0	5	22	89	72	24	1	0	0
	CDH23.3	929	0	0	0	0	22	83	445	292	86	1	0	0
	CDH26.7	115	0	0	0	0	2	7	73	27	6	0	0	0
Wind	WSAvg	4.2	4.6	4.5	4.4	4.5	3.7	3.5	3.6	3.7	3.9	4.6	4.7	4.7
Precipitation	PrecAvg	1003	74	67	80	82	81	77	63	86	97	89	102	104
	PrecMax	1222	190	138	143	158	138	136	156	198	185	146	156	211
	PrecMin	795	25	16	33	33	34	29	17	32	29	26	51	48
	PrecStd	107	40	37	30	37	30	33	35	37	44	32	30	42
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	7.1	7.9	15.1	19.1	25.5	27.7	31.3	29.1	27.5	21.4	15.2	11.3
		MCWB	6.4	5.5	12.5	13.3	17.8	20.0	23.1	21.8	22.3	17.7	12.4	9.9
	2%	DB	5.2	5.6	10.3	16.1	23.5	25.6	28.6	27.1	25.5	19.6	13.3	8.6
		MCWB	4.3	4.0	6.8	10.2	16.9	19.3	22.1	21.5	21.5	16.8	10.8	7.3
	5%	DB	3.6	3.8	7.9	13.7	21.5	23.9	26.9	25.9	24.1	18.0	12.0	7.0
		MCWB	2.4	2.2	5.4	8.7	16.2	18.6	21.5	21.3	20.9	15.3	9.9	6.0
	10%	DB	2.2	2.5	6.0	11.8	19.6	22.5	25.7	25.0	22.8	16.6	10.6	5.4
		MCWB	1.0	1.1	3.9	8.0	14.9	17.7	20.9	20.8	19.9	14.2	8.6	4.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	6.5	6.3	12.3	14.6	20.1	22.7	25.2	24.2	23.9	19.2	13.1	10.0
		MCDB	6.9	7.1	14.2	17.8	23.5	25.8	28.1	27.2	26.4	20.4	15.0	11.0
	2%	WB	4.6	4.3	7.8	11.7	18.4	20.9	23.8	23.3	22.5	17.2	11.3	7.5
		MCDB	5.1	5.6	9.4	14.2	21.8	23.8	26.6	25.4	24.5	18.8	12.9	8.4
	5%	WB	2.6	2.5	5.6	10.0	17.1	19.7	22.9	22.6	21.5	15.9	10.1	5.9
		MCDB	3.4	3.6	7.4	12.4	20.2	22.5	25.4	24.7	23.5	17.7	11.7	6.9
	10%	WB	1.1	1.2	4.0	8.4	15.7	18.7	22.0	21.8	20.4	14.3	8.9	4.1
		MCDB	2.1	2.2	5.9	11.0	18.9	21.2	24.7	24.0	22.5	16.3	10.6	5.3
Mean Daily Temperature Range	5% DB	MDBR	8.2	7.9	7.7	8.6	9.4	8.5	8.5	8.3	9.0	7.8	7.4	6.7
		MCDBR	7.5	7.8	9.4	12.0	11.7	10.5	9.6	8.8	9.0	8.3	8.8	7.5
	5% WB	MCWBR	7.2	7.0	7.1	7.9	7.0	6.2	5.6	5.3	6.1	6.3	7.4	7.4
		MCWBR	7.2	7.8	8.5	10.1	9.9	8.4	8.2	7.1	8.1	7.5	7.8	7.5
Clear-Sky Solar Irradiance	taub		0.292	0.297	0.322	0.355	0.383	0.421	0.426	0.417	0.372	0.350	0.308	0.284
		taud	2.407	2.394	2.431	2.397	2.372	2.288	2.282	2.304	2.412	2.463	2.525	2.493
	Ebn at Noon		833	897	917	908	888	851	842	837	851	824	814	814
		Edh at Noon	74	90	99	112	119	130	129	122	101	84	67	63
All-Sky Solar Radiation	RadAvg		1.29	2.15	3.49	4.48	5.44	5.78	5.89	5.18	4.13	2.46	1.52	1.03
	RadStd		0.13	0.20	0.30	0.50	0.53	0.46	0.49	0.26	0.34	0.22	0.18	0.12

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (32 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page